

Nirma University

Institute of Technology

Semester End Examination (IR), May-2026

Master of Technology in Computer Science & Engineering (Data Science),

Semester- II

6CS268ME25 Soft Computing

Roll/

Exam No.

25MCD005

Supervisor's Initial
with date

6/5

Time : 3 Hours

Max Marks: 100

Instructions:

1. All questions are compulsory. (No optional questions)
2. Use section-wise separate answer book.
3. Figures to right indicate full marks.
4. Draw neat sketches whenever necessary.
5. Assume suitable data wherever necessary and state your assumptions clearly.

SECTION-I

Q-1 Answer the following questions.

[20]

A
CO2
BL3

Consider the 8-puzzle problem with the following Initial State (S_{init}) and Goal State (S_{goal}):

$$S_{init} = \begin{bmatrix} 1 & 3 \\ 8 & 2 & 4 \\ 7 & 6 & 5 \end{bmatrix}, \quad S_{goal} = \begin{bmatrix} 1 & 2 & 3 \\ 8 & & 4 \\ 7 & 6 & 5 \end{bmatrix}$$

Solve this problem from S_{init} to S_{goal} using the Simple Hill Climbing algorithm.

- i. Use the Manhattan Distance heuristic (h), strictly ignoring the blank tile (the empty space) in your calculations.
- ii. Assume the algorithm generates successors by moving the blank tile in the following strict order: UP, DOWN, LEFT, RIGHT.
- iii. Show the heuristic calculation for every generated node. Clearly indicate when the algorithm accepts a state and why it stops evaluating the remaining neighbors at that level.

B
CO3
BL4

Write the standard Generate-and-Test algorithmic steps. Critically analyze its primary strengths and weaknesses in the context of solving state-space search problems.

8

Q-2 Answer the following questions.

[20]

A
CO3
BL4

Assume you are designing a Genetic Algorithm to solve a 10-city symmetric Traveling Salesman Problem (TSP). Explain why standard crossover operators (such as single-point or uniform crossover) are unsuitable for this problem. Identify and explain in detail any three specialized permutation-based crossover operators that can successfully maintain valid tours. Demonstrate step-by-step execution of these operators using two explicit 10-city parent chromosomes to produce at least one valid offspring.

12

B
CO1
BL2

Contrast Crisp Sets and Fuzzy Sets by providing at least four distinct conceptual differences. Support your comparison by defining a real-world example and explaining exactly how it would be evaluated differently in a crisp system versus a fuzzy system.

8

Q-3
CO2
BL3

6CS268ME25 SOL
Consider a dataset with three 1-dimensional data points: $x_1 = 2$, $x_2 = 4$, and $x_3 = 9$. You are required to group this data into two clusters ($c=2$) using the Fuzzy C-Means (FCM) clustering algorithm with a fuzzifier value of $m = 2$. Assume the initial randomly generated membership matrix U (where rows represent clusters and columns represent data points) is:

$$U = \begin{bmatrix} 0.8 & 0.6 & 0.2 \\ 0.2 & 0.4 & 0.8 \end{bmatrix}$$

- (i) Write the mathematical formula for calculating the cluster centers. (ii) Calculate the precise values for the new cluster centers, v_1 and v_2 . (iii) Using the newly calculated cluster centers, compute the updated membership value for the first data point x_1 belonging to the first cluster (μ_{11}).

SECTION-II

Q-4 Answer the following questions.

A
CO2
BL3

Write the complete algorithmic steps for the Roulette Wheel Selection (Fitness Proportionate Selection) method used in Genetic Algorithms. Demonstrate its step-by-step execution to select exactly two parents from a population of four individuals with the following fitness values: $f_1 = 10$, $f_2 = 20$, $f_3 = 30$, and $f_4 = 40$. Assume these are scaled fitness values for a maximization problem, where a higher value indicates a better solution. Assume the uniformly generated random numbers for the two selections are $r_1 = 0.45$ and $r_2 = 0.85$. Show all calculations for selection probability, cumulative probability, and the final selected mating pool.

B
CO1
BL2

Outline the standard algorithmic execution of Differential Evolution (DE). Specifically, explain the four primary operational phases—Initialization, Mutation, Crossover (Recombination), and Selection—detailing how the target vector, mutant vector, and trial vector are generated and interact during these phases.

Q-5 Answer the following questions.

A
CO3
BL4

The Ant Colony Optimization (ACO) algorithm is biologically inspired by the foraging behavior of natural ant colonies seeking the shortest path to a food source. Outline the step-by-step algorithmic execution of a standard ACO applied to a routing problem (such as the Traveling Salesman Problem). Critically analyze the impact of the following three control parameters on the algorithm's balance between exploration and exploitation: (i) The pheromone exponential weight (α) (ii) The heuristic visibility exponential weight (β).

B
CO4
BL5

In Ant Colony Optimization (ACO), the pheromone evaporation rate (ρ) is used to update the trails at the end of each iteration: $\tau = (1 - \rho) \times \tau$. Consider two extreme configurations for a routing problem: Configuration 1: The evaporation rate is set extremely low ($\rho = 0.01$). Configuration 2: The evaporation rate is set extremely high ($\rho = 0.99$). Critically evaluate the negative impact each configuration would have on the swarm's ability to find the optimal path. Based on this, justify the statement: "Pheromone evaporation is the primary mechanism for maintaining exploration in an artificial ant colony."

Q-6
CO4

While traditional Particle Swarm Optimization (PSO) is used for mathematical functions, it can be adapted to solve unsupervised

BL6

clustering. Formulate a PSO model to group a dataset of 2-dimensional data points ($D=2$, representing x, y coordinates) into exactly 3 distinct clusters ($K=3$). In your design, you must explicitly define: (i) **Particle Representation**: How exactly is a single particle encoded? What is its exact total length? Provide a visual array representation for this specific scenario ($K=3, D=2$). (ii) **Fitness Function**: Formulate the mathematical objective function used to evaluate a particle. What specific metric is being minimized? (iii) **Mechanism**: Explain how the continuous velocity and position update equations physically manipulate the cluster centroids in the 2D data space during an iteration.

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